APPENDIX II. UPDATE TO THE LOW-INCOME COUNTRY DEBT SUSTAINABILITY ANALYSIS

This annex provides an update to the LIC-DSA attached to the completion point document of September 12, 2012 (IMF Country Report No. 12/295*). It includes updated macroeconomic projections and additional information on borrowing, including on four nonconcessional loans contracted in 2012 and early in 2013.*

1. Guinea continues to be assessed at a moderate risk of debt distress. As in the previous assessment, the moderate rating is primarily due to Guinea's vulnerability to adverse shocks to growth, exports, foreign direct investment flows, and fiscal performance. This underscores the need for sustained implementation of sound macroeconomic policies, ensuring public investment in sound projects, while improving implementation capacity and the business climate; the above referenced previous LIC-DSA provides a more elaborate discussion of these issues. In addition, especially with the projected expansion of the export-oriented mining sector, structural reforms aimed at removing bottlenecks and improving productivity in the economy to minimize the effects (Dutch disease) on the competitiveness on other sectors of the economy. Guinea's vulnerabilities indicate that new borrowings on a large scale should be avoided, especially before the boost to exports and growth from the onset of new mining production. As noted in the previous DSA, the authorities continue to be committed to ensuring that government participation in large-scale infrastructure and mining projects does not involve the use of budgetary resources to meet the associated financing needs.

2. Guinea's external public debt amounted to $1.3 billion (23 percent of GDP) at end-2012 (Table 1), significantly down following the attainment of the HIPC completion point from $3.2 billion (62 percent of GDP) at end-2011, and slightly above that projected in the previous LIC-DSA ($1.1 million). The underlying macroeconomic framework is consistent with the ECF-supported program, but differs from the previous LIC-DSA (Table 2). First, the government's public investment program, which was finalized in late 2012, has been incorporated in the budget projections. This has raised the level of public investment, in particular because of the inclusion of a major investment project, the Kaleta hydroelectric dam. Second, delays in investment and exports related to the large Simandou iron ore project have resulted in a (downward) revision in the projections of real GDP growth, exports, and fiscal revenues—a substantial increase in these three variables, which had been projected in 2015, has now been shifted to 2017. In addition, the discount rate used for calculating present values has been reduced from 4 percent to 3 percent, resulting in an increase in the present value of debt. Debt policy thresholds are unchanged.

3. The results of the external LIC-DSA confirm that Guinea's debt dynamics are sustainable. The stress tests reaffirm that Guinea continues to have vulnerabilities primarily from shocks to growth, exports, foreign direct investment flows, and fiscal performance. These vulnerabilities to the alternative scenarios and stress tests highlight the dependence of the projected improvement in macroeconomic prospects under the baseline on the projected large inflow of foreign direct investments in the mining sector and the related jump in mining exports. The main changes to the baseline compared to the previous LIC-DSA reflects a lower grant element of new borrowing in the latter part of the projection period, the higher present value (PV) of debt, lower

debt service payments in 2013–16, and the delayed start-up of operations related to the Simandou mining project and its impact on real GDP growth, exports and fiscal revenue (Figure 1). The lower grant element of new borrowing is primarily because, compared with the last LIC-DSA, the current one allows for some financing from commercial creditors beginning 2027, the introduction of specific terms of four new nonconcessional loans, in particular that for the Kaleta hydroelectric dam, and the increase in the discount rate used to calculate the grant element. The higher upfront PV of debt stems from the higher assumed borrowing (and corresponding government disbursements) to finance the revised investment budget, which is associated with an increase in external borrowing during 2013–16 (Figures 1 and 2, Tables 3 and 4). The bulk of the increased borrowing is accounted for by a loan from the Export-Import Bank of China to finance the construction of the Kaleta hydroelectric dam; the loan amount is $335 million (5.3 percent of 2013 GDP) and has a grant element of 33.3 percent. This loan and the three other nonconcessional loans are projected to be disbursed during 2013–16. The changes in the debt service ratios reflect the higher borrowing, and in the short-term a change in the assumed profile of debt service payments on French ODA claims that have been converted into debt-for-development swaps (C2D); these payments are assumed in the present LIC-DSA to be stretched out over 2013–20 rather than over 2013–15 as projected in the last LIC-DSA.[1] The average interest rate on new borrowing is relatively low (similar to the last LIC-DSA), reflecting the presence of several long-standing lenders, from whom further lending is expected, where the loan terms have a zero interest rate, and for others (even for the four new nonconcessional loans) interest rates charged have been low. While similar terms are projected for this group of creditors, there is a risk that interest rates will rise as the global economy recovers from the recent economic and financial crisis. The delay in the Simandou mining project results in a shift of about two years in the peaks in the debt and debt service ratios; this is highlighted in the most extreme shock stress test.[2] All other baseline assumptions are broadly similar to the last LIC-DSA.

4. The inclusion of domestic debt in the LIC-DSA worsens the debt burden indicators, **although the domestic debt burden is expected to decrease over time (Figure 3**, **Tables 5 and 6)**. Following the large increase in borrowing from the domestic banking system in 2009–10, the authorities, in line with their policies to restore macroeconomic stability, have sharply curtailed new borrowing, and net repayments of domestic debt are expected to resume.[3] As in the last LIC-DSA, the public debt position is vulnerable notably with respect to policy reversals in maintaining fiscal

[1] Under C2D (*Contrats de désendettement et développement*) the debt service maturities at the time of the HIPC completion point were consolidated and re-profiled; when they are repaid, they are returned to the country through grants from France for development spending. At the time of staff discussions in the preparation of the HIPC completion point documents, the assumed time period over which the C2D would be implemented was based on preliminary information. As a result of discussion after the completion point between the authorities and the French Development Agency, the agreed re-profiling was longer than previously assumed to take into account the absorptive capacity of Guinea.

[2] The most extreme stress test corresponds to the combination shock.

[3] In addition, under an agreement between the Ministry of Finance and the central bank, the government will repay past advances through end-2010 from the central bank, over a period of 40 years, starting in 2020. The agreement envisages that the repayments will be made in cash rather than issuance of government debt. However, discussions are to begin in the near future on the recapitalization of the central bank which would likely entail the issuance of new public debt and would offset to some extent the projected decline of the share of domestic debt in total public sector debt.

discipline and shocks to growth. The impact of an unchanged primary balance is much weaker than in the previous LIC-DSA because the reference year changed from 2012 to 2013 and the projected 2013 primary deficit is smaller than the 2012 deficit projected at the time of the last LIC-DSA.

5. The Guinean authorities broadly concurred with the assumptions and conclusions, including identified vulnerabilities, of this update of the LIC-DSA.

Table 1. Guinea: Structure of External Public Debt (end-2012, nominal)

	$ Million	Percent of total	Percent of GDP
Total	**1,306.3**	**100.0**	**23.2**
Multilateral creditors	**585.5**	44.8	10.4
IMF	57.3	4.4	1.0
World Bank	155.2	11.9	2.8
AfDB Group	96.6	7.4	1.7
IsDB	128.8	9.9	2.3
EU	0.0	0.0	0.0
Other multilateral creditors	147.6	11.3	2.6
Official bilateral creditors	**659.9**	**50.5**	**11.7**
Paris Club	272.0	20.8	4.8
Non-Paris Club	263.5	20.2	4.7
Arab Funds	124.4	9.5	2.2
Commercial creditors	**60.8**	**4.7**	**1.1**

Sources: Guinean authorities, and World Bank and IMF staff estimates.

Table 2. Guinea's LIC DSA Macroeconomic Assumptions: Comparison with the HIPC Completion Point LIC DSA

	HICP Completion Point LIC DSA						Current LIC DSA Update					
	2013	2014	2015	2020	2025	2032	2013	2014	2015	2020	2025	2032
	(percent of GDP, unless otherwise indicated)											
Nominal GDP ($ million)	6,237.1	6,179.7	7,475.2	13,396.2	16,008.1	21,388.3	6,273.0	6,347.9	6,797.7	14,092.3	17,314.9	24,296.4
Real GDP (percentage change)	5.0	5.2	19.9	6.1	2.7	3.1	4.5	5.2	5.4	6.4	3.2	3.6
Fiscal (central government)												
Revenue (incl. external grants)	24.0	24.9	27.5	30.2	29.8	32.3	22.7	24.0	23.6	25.9	26.5	26.9
Primary expenditure	25.2	24.9	27.1	31.4	29.2	31.3	26.8	25.5	24.1	26.6	26.0	26.1
of which Capital expenditure and net lending	10.3	10.5	13.2	18.8	18.6	23.3	12.1	10.8	9.4	13.0	13.9	17.4
Primary fiscal balance	-1.2	0.0	0.4	-1.2	0.6	0.9	-4.1	-1.5	-0.6	-0.7	0.5	0.8
Basic primary fiscal balance	-0.6	0.4	0.4	1.4	1.3	1.1	-2.4	0.2	0.3	1.4	0.8	0.6
New external borrowing[1]	1.6	1.5	1.0	1.7	0.5	0.1	4.9	3.1	1.7	1.6	0.5	0.1
Grant element of new external borrowing (%)	39.7	39.7	39.7	33.4	33.4	30.3	17.7	15.5	20.3	23.0	23.0	17.4
Balance of payments												
Exports of goods and services	26.4	28.2	41.4	64.5	56.6	43.9	26.9	28.0	26.9	52.6	51.5	37.5
Imports of goods and services	-63.5	-65.2	-61.1	-52.3	-47.2	-40.4	-51.7	-73.1	-67.8	-49.0	-46.8	-40.3
Current account, incl. official transfers	-39.2	-39.2	-21.3	-2.0	-2.7	-5.8	-25.2	-46.9	-42.1	-7.0	-5.8	-10.5
Foreign direct investment	35.8	35.9	25.7	3.7	4.1	4.7	14.8	41.8	35.8	3.6	3.9	4.6

Source: Country authorities, IMF and World staffs.
[1] Includes publicly guaranteed external borrowing.

Figure 1. Guinea's External LIC DSA Comparison with the HIPC Completion Point LIC DSA

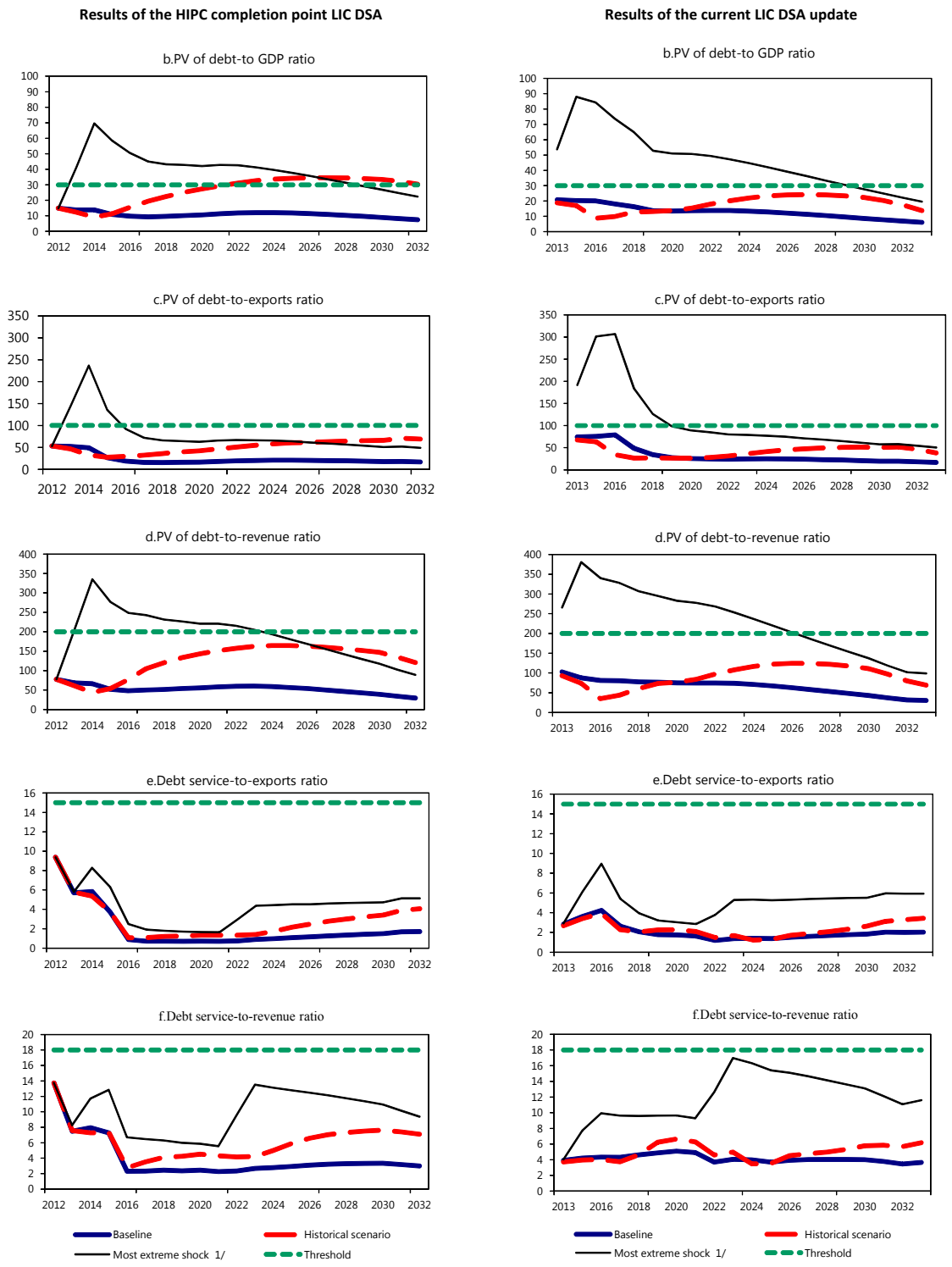
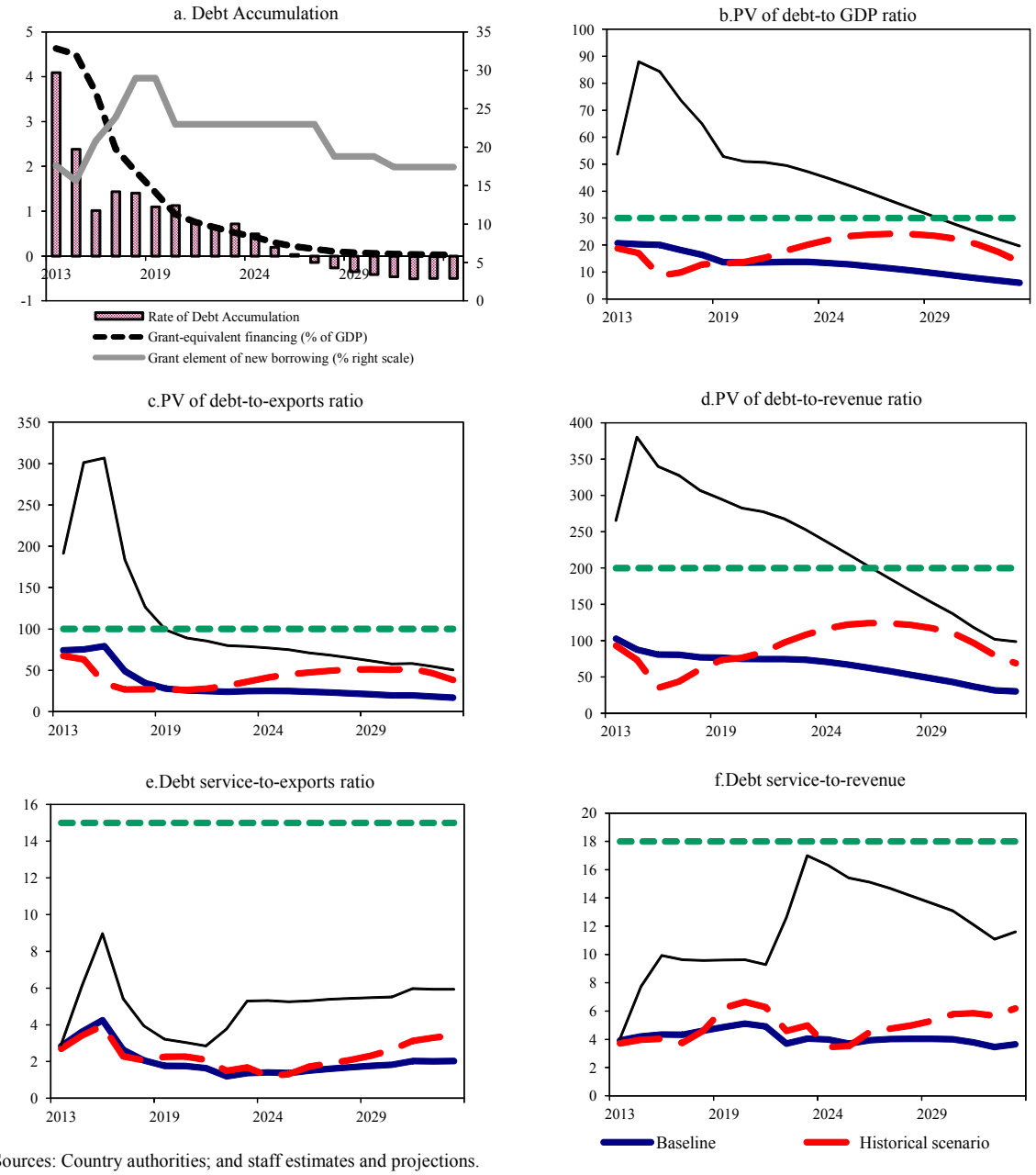


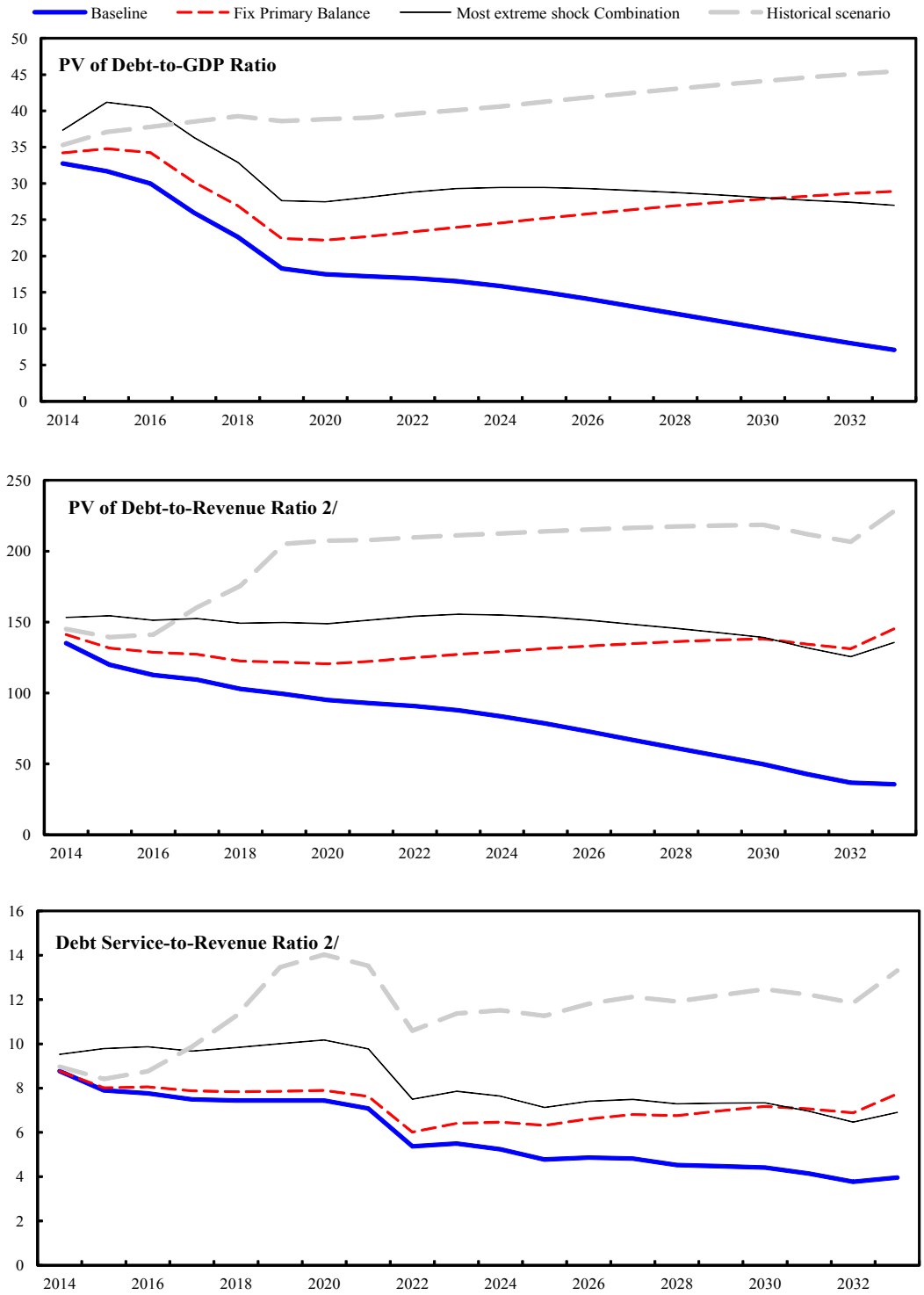
Figure 2. Guinea: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2013–33 [1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2023. In figure b. it corresponds to a Combination shock; in c. to a Combination shock; in d. to a Combination shock; in e. to a Combination shock and in figure f. to a Combination shock

Figure 3. Guinea: Indicators of Public Debt Under Alternative Scenarios, 2013–33 [1]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2023.
2/ Revenues are defined inclusive of grants.

Table 3. Guinea: External Debt Sustainability Framework, Baseline Scenario, 2010–33 [1]

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2010	2011	2012			2013	2014	2015	2016	2017	2018	2013-2018 Average	2023	2033	2019-2033 Average
External debt (nominal) 1/	**70.7**	**67.5**	**23.0**			**26.1**	**28.1**	**27.1**	**26.7**	**24.2**	**21.9**		**18.2**	**8.3**	
o/w public and publicly guaranteed (PPG)	70.7	67.5	23.0			26.1	28.1	27.1	26.7	24.2	21.9		18.2	8.3	
Change in external debt	-1.0	-3.2	-44.5			3.1	2.0	-0.9	-0.4	-2.6	-2.3		-0.1	-1.0	
Identified net debt-creating flows	4.8	11.4	17.6			9.4	3.8	5.0	13.4	-0.7	-2.7		0.7	6.6	
Non-interest current account deficit	**10.8**	**19.6**	**33.4**	**9.3**	**10.6**	**24.9**	**46.7**	**41.9**	**39.3**	**18.9**	**4.4**		**4.8**	**11.5**	7.1
Deficit in balance of goods and services	7.4	22.2	32.3			24.8	45.1	40.9	38.3	17.9	3.2		-6.4	4.3	
Exports	29.1	31.6	29.7			26.9	28.0	26.9	25.3	36.9	47.4		55.2	35.8	
Imports	36.5	53.8	62.0			51.7	73.1	67.8	63.7	54.8	50.7		48.8	40.2	
Net current transfers (negative = inflow)	-5.1	-10.5	-6.0	-6.9	1.9	-6.3	-5.0	-5.2	-4.9	-4.2	-3.6		-2.5	-1.8	-2.3
o/w official	0.0	-2.2	-1.0			-1.7	-0.4	-0.4	-0.4	-0.4	-0.3		-0.2	-0.3	
Other current account flows (negative = net inflow)	8.5	7.9	7.2			6.5	6.5	6.2	5.9	5.3	4.8		13.7	8.9	
Net FDI (negative = inflow)	**-2.4**	**-6.5**	**-10.4**	**-5.1**	**2.9**	**-14.8**	**-41.8**	**-35.8**	**-24.7**	**-15.6**	**-3.6**		**-3.8**	**-4.6**	-4.0
Endogenous debt dynamics 2/	**-3.6**	**-1.7**	**-5.4**			**-0.7**	**-1.1**	**-1.1**	**-1.2**	**-4.0**	**-3.5**		**-0.3**	**-0.2**	
Contribution from nominal interest rate	0.7	1.0	0.7			0.2	0.3	0.3	0.3	0.3	0.3		0.3	0.1	
Contribution from real GDP growth	-1.3	-2.7	-2.4			-0.9	-1.3	-1.4	-1.5	-4.3	-3.7		-0.5	-0.3	
Contribution from price and exchange rate changes	-3.0	0.0	-3.7			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**-5.8**	**-14.6**	**-62.1**			**-6.3**	**-1.8**	**-5.9**	**-13.8**	**-1.8**	**0.4**		**-0.9**	**-7.6**	
o/w exceptional financing	-0.4	-1.6	-39.9			-2.1	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	…	…	37.1			18.8	20.8	20.3	20.1	18.1	16.3		13.7	6.0	
In percent of exports	…	…	122.4			69.9	74.3	75.4	79.2	49.1	34.5		24.9	16.8	
PV of PPG external debt	**…**	**…**	**37.1**			**18.8**	**20.8**	**20.3**	**20.1**	**18.1**	**16.3**		**13.7**	**6.0**	
In percent of exports	**…**	**…**	**122.4**			**69.9**	**74.3**	**75.4**	**79.2**	**49.1**	**34.5**		**24.9**	**16.8**	
In percent of government revenues	**…**	**…**	**220.8**			**94.0**	**102.8**	**87.7**	**81.0**	**80.5**	**77.2**		**73.6**	**30.3**	
Debt service-to-exports ratio (in percent)	**10.1**	**13.8**	**11.6**			**3.0**	**2.8**	**3.6**	**4.2**	**2.6**	**2.1**		**1.4**	**2.0**	
PPG debt service-to-exports ratio (in percent)	**10.1**	**13.8**	**11.6**			**3.0**	**2.8**	**3.6**	**4.2**	**2.6**	**2.1**		**1.4**	**2.0**	
PPG debt service-to-revenue ratio (in percent)	**19.2**	**26.0**	**17.1**			**4.1**	**3.9**	**4.2**	**4.3**	**4.3**	**4.6**		**4.1**	**3.7**	
Total gross financing need (Billions of U.S. dollars)	0.6	0.9	1.5			0.7	0.4	0.5	1.2	0.4	0.2		0.3	1.9	
Non-interest current account deficit that stabilizes debt ratio	11.8	22.8	77.9			21.9	44.7	42.8	39.7	21.5	6.6		5.0	12.5	
Key macroeconomic assumptions															
Real GDP growth (in percent)	1.9	3.9	3.9	2.5	1.5	4.5	5.2	5.4	5.8	18.9	18.2	9.7	3.0	3.7	4.8
GDP deflator in US dollar terms (change in percent)	4.3	0.0	5.7	4.2	15.3	6.6	-3.8	1.6	2.3	-0.1	-0.5	1.0	1.2	1.7	1.5
Effective interest rate (percent) 5/	1.1	1.4	1.2	1.4	0.3	1.0	1.1	1.1	1.2	1.2	1.3	1.2	1.4	1.4	1.4
Growth of exports of G&S (US dollar terms, in percent)	11.4	12.9	3.3	8.6	12.8	1.0	5.0	3.1	2.0	73.0	51.2	22.5	0.8	0.7	4.6
Growth of imports of G&S (US dollar terms, in percent)	26.2	53.2	26.6	16.6	19.3	-7.1	43.1	-0.7	1.7	2.1	8.8	8.0	2.0	5.1	4.7
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	17.6	15.6	20.8	23.9	29.0	29.0	22.7	23.0	17.4	20.7
Government revenues (excluding grants, in percent of GDP)	15.3	16.8	20.1			20.0	20.2	23.1	24.8	22.5	21.2		18.7	19.9	19.4
Aid flows (in Billions of US dollars) 7/	0.0	0.2	0.2			0.3	0.3	0.3	0.2	0.2	0.2		0.1	0.0	
o/w Grants	0.0	0.2	0.2			0.2	0.3	0.2	0.1	0.1	0.1		0.0	0.0	
o/w Concessional loans	0.0	0.0	0.0			0.1	0.0	0.0	0.1	0.1	0.1		0.1	0.0	
Grant-equivalent financing (in percent of GDP) 8/	…	…	…			4.6	4.5	3.6	2.4	1.9	1.4		0.4	0.0	0.3
Grant-equivalent financing (in percent of external financing) 8/	…	…	…			53.5	63.3	72.9	56.3	53.3	49.4		33.6	29.9	34.7
Memorandum items:															
Nominal GDP (Billions of US dollars)	4.9	5.1	5.6			6.3	6.3	6.8	7.4	8.7	10.3		15.9	25.6	
Nominal dollar GDP growth	6.3	4.0	9.9			11.4	1.2	7.1	8.3	18.7	17.6	10.7	4.2	5.4	6.4
PV of PPG external debt (in Billions of US dollars)			1.8			1.1	1.3	1.4	1.4	1.6	1.6		2.1	1.5	
(PVt-PVt-1)/GDPt-1 (in percent)						4.1	2.4	1.0	1.4	1.4	1.1	1.9	0.5	-0.5	0.1
Gross workers' remittances (Billions of US dollars)	…	…	…			…	…	…	…	…	…		…	…	
PV of PPG external debt (in percent of GDP + remittances)	…	…	37.1			18.8	20.8	20.3	20.1	18.1	16.3		13.7	6.0	
PV of PPG external debt (in percent of exports + remittances)	…	…	122.4			69.9	74.3	75.4	79.2	49.1	34.5		24.9	16.8	
Debt service of PPG external debt (in percent of exports + remittances)	…	…	11.6			3.0	2.8	3.6	4.2	2.6	2.1		1.4	2.0	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 4. Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–33

(In percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to GDP ratio								
Baseline	19	21	20	20	18	16	**14**	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	19	19	17	9	10	13	**20**	14
A2. New public sector loans on less favorable terms in 2013-2033 2	19	21	21	21	20	19	**18**	10
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	19	21	22	21	19	17	**15**	6
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	19	22	25	25	22	20	**16**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	19	22	25	24	22	20	**17**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	19	53	80	76	67	59	**43**	18
B5. Combination of B1-B4 using one-half standard deviation shocks	19	54	88	84	74	65	**47**	20
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	19	29	29	29	26	23	**20**	9
PV of debt-to-exports ratio								
Baseline	70	74	75	79	49	34	**25**	17
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	70	67	63	34	27	27	**37**	38
A2. New public sector loans on less favorable terms in 2013-2033 2	70	75	78	85	54	40	**32**	29
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	70	72	74	78	48	34	**24**	16
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	70	87	110	115	71	49	**34**	23
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	70	72	74	78	48	34	**24**	16
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	70	191	296	301	181	124	**77**	50
B5. Combination of B1-B4 using one-half standard deviation shocks	70	192	301	307	184	126	**79**	51
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	70	72	74	78	48	34	**24**	16
PV of debt-to-revenue ratio								
Baseline	94	103	88	81	80	77	**74**	30
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	94	93	74	35	44	61	**108**	69
A2. New public sector loans on less favorable terms in 2013-2033 2	94	104	91	86	89	89	**94**	51
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	94	104	93	86	86	82	**78**	32
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	94	110	109	100	98	94	**86**	35
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	94	109	106	98	98	94	**89**	37
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	94	263	345	308	297	278	**229**	89
B5. Combination of B1-B4 using one-half standard deviation shocks	94	266	381	340	328	307	**253**	99
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	94	146	125	115	115	110	**105**	43

Table 4. Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–33 (continued)

(In percent)

Debt service-to-exports ratio								
Baseline	3	3	4	4	3	2	**1**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	3	3	3	4	2	2	**2**	3
A2. New public sector loans on less favorable terms in 2013-2033 2	3	3	4	4	3	2	**2**	2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	3	3	4	4	3	2	**1**	2
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	3	3	4	5	3	3	**2**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	3	3	4	4	3	2	**1**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	3	3	6	9	5	4	**5**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	3	3	6	9	5	4	**5**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	3	3	4	4	3	2	**1**	2
Debt service-to-revenue ratio								
Baseline	4	4	4	4	4	5	**4**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	4	4	4	4	4	5	**5**	6
A2. New public sector loans on less favorable terms in 2013-2033 2	4	4	4	4	5	5	**5**	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	4	4	5	5	5	5	**4**	4
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	4	4	4	5	5	5	**5**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	4	4	5	5	5	6	**5**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	4	4	7	9	9	9	**15**	10
B5. Combination of B1-B4 using one-half standard deviation shocks	4	4	8	10	10	10	**17**	12
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	4	6	6	6	6	7	**6**	5
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	17	17	17	17	17	17	**17**	17

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 5. Guinea: Public Sector Debt Sustainability Framework, Baseline Scenarios, 2010–33

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2010	2011	2012	Average 5/	Standard Deviation 5/	2013	2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
Public sector debt 1/	99.6	77.8	35.4			39.2	40.0	38.5	36.6	32.0	28.2		21.0	9.4	
o/w foreign-currency denominated	70.7	67.5	23.0			26.1	28.1	27.1	26.7	24.2	21.9		18.2	8.3	
Change in public sector debt	10.3	-21.9	-42.3			3.8	0.8	-1.5	-1.9	-4.6	-3.9		-0.5	-1.1	
Identified debt-creating flows	10.9	-10.4	-42.4			-0.3	-1.3	-2.5	-2.3	-4.3	-3.5		-0.3	-1.0	
Primary deficit	12.0	-0.7	1.6	1.9	4.4	1.6	-0.2	-0.5	0.0	0.9	0.8	0.4	0.2	-0.6	-0.2
Revenue and grants	15.7	20.2	22.9			23.8	24.2	26.4	26.6	23.7	22.0		18.8	19.9	
of which: grants	0.4	3.4	2.7			3.8	4.0	3.3	1.8	1.2	0.8		0.2	0.0	
Primary (noninterest) expenditure	27.7	19.5	24.4			25.4	24.0	25.9	26.6	24.6	22.8		19.1	19.3	
Automatic debt dynamics	-0.6	-8.1	-10.8			-1.9	-1.2	-2.1	-2.3	-5.2	-4.3		-0.6	-0.4	
Contribution from interest rate/growth differential	-3.5	-7.7	-3.4			-1.9	-2.2	-2.0	-2.2	-5.7	-4.8		-0.7	-0.4	
of which: contribution from average real interest rate	-1.8	-4.0	-0.5			-0.3	-0.3	0.1	0.0	0.2	0.1		-0.1	-0.1	
of which: contribution from real GDP growth	-1.7	-3.7	-3.0			-1.5	-1.9	-2.0	-2.1	-5.8	-4.9		-0.6	-0.4	
Contribution from real exchange rate depreciation	2.9	-0.4	-7.3			-0.1	1.1	-0.1	-0.1	0.5	0.5		...	...	
Other identified debt-creating flows	-0.5	-1.6	-33.2			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-0.5	-1.6	-33.2			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-0.5	-11.4	0.0			4.1	2.2	1.0	0.4	-0.3	-0.4		-0.2	-0.1	
Other Sustainability Indicators															
PV of public sector debt	...	10.2	48.0			31.9	32.8	31.7	30.0	26.0	22.6		16.5	7.1	
o/w foreign-currency denominated	...	0.0	37.1			18.8	20.8	20.3	20.1	18.1	16.3		13.7	6.0	
o/w external	...	...	37.1			18.8	20.8	20.3	20.1	18.1	16.3		13.7	6.0	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	16.7	4.8	6.0			3.6	1.9	1.6	2.1	2.7	2.5		1.3	0.2	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	50.7	210.1			134.2	135.2	120.0	112.8	109.5	102.9		87.9	35.6	
PV of public sector debt-to-revenue ratio (in percent)	…	61.1	238.3			159.4	162.1	137.1	121.0	115.4	106.9		88.7	35.6	
o/w external 3/	…	…	184.2			94.0	102.8	87.7	81.0	80.5	77.2		73.6	30.3	
Debt service-to-revenue and grants ratio (in percent) 4/	30.0	27.1	19.6			8.5	8.8	7.9	7.8	7.5	7.4		5.5	4.0	
Debt service-to-revenue ratio (in percent) 4/	30.7	32.6	22.2			10.1	10.5	9.0	8.3	7.9	7.7		5.6	4.0	
Primary deficit that stabilizes the debt-to-GDP ratio	1.7	21.2	43.9			-2.2	-1.0	1.0	1.9	5.5	4.7		0.7	0.5	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	1.9	3.9	3.9	2.5	1.5	4.5	5.2	5.4	5.8	18.9	18.2	9.7	3.0	3.7	4.8
Average nominal interest rate on forex debt (in percent)	1.1	1.4	1.2	1.4	0.3	1.0	1.1	1.1	1.2	1.2	1.3	1.2	1.4	1.4	1.4
Average real interest rate on domestic debt (in percent)	-9.2	-12.5	-1.1	-5.2	7.2	-2.1	-1.1	2.1	1.2	3.9	4.2	1.4	0.6	0.4	0.8
Real exchange rate depreciation (in percent, + indicates depreciation)	4.1	-0.5	-11.4	0.7	19.7	-0.3	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	20.2	19.7	12.9	17.5	9.4	9.9	9.6	6.3	6.5	3.8	3.4	6.6	5.1	5.6	5.4
Growth of real primary spending (deflated by GDP deflator, in percent)	0.3	-0.3	0.3	0.1	0.2	0.1	0.0	0.1	0.1	0.1	0.1	0.1	0.0	-0.1	0.0
Grant element of new external borrowing (in percent)	...	...	0.0	0.0	…	17.6	15.6	20.8	23.9	29.0	29.0	22.7	23.0	17.4	...

Sources: Country authorities; and staff estimates and projections.

1/ Public sector covers central government; net debt is used.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 6. Guinea: Sensivity Analysis for Key Indicators of Public Debt, 2013–33

	Projections							
	2013	2014	2015	2016	2017	2018	2023	2033
PV of Debt-to-GDP Ratio								
Baseline	32	33	32	30	26	23	17	7
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	32	35	37	38	39	39	40	45
A2. Primary balance is unchanged from 2013	32	34	35	34	30	27	24	29
A3. Permanently lower GDP growth 1/	32	33	32	31	27	24	19	17
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	32	35	37	37	33	31	29	29
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	32	38	42	40	35	30	22	11
B3. Combination of B1-B2 using one half standard deviation shocks	32	37	41	40	36	33	29	27
B4. One-time 30 percent real depreciation in 2014	32	42	41	38	33	28	21	12
B5. 10 percent of GDP increase in other debt-creating flows in 2014	32	41	40	37	32	28	21	10
PV of Debt-to-Revenue Ratio 2/								
Baseline	134	135	120	113	109	103	88	36
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	134	145	139	141	160	176	211	228
A2. Primary balance is unchanged from 2013	134	141	132	129	127	123	127	145
A3. Permanently lower GDP growth 1/	134	136	121	115	113	108	103	85
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	134	143	138	137	140	138	152	145
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	134	157	160	151	146	138	118	57
B3. Combination of B1-B2 using one half standard deviation shocks	134	153	155	151	153	149	156	136
B4. One-time 30 percent real depreciation in 2014	134	174	153	143	138	129	112	62
B5. 10 percent of GDP increase in other debt-creating flows in 2014	134	169	150	141	137	129	110	51
Debt Service-to-Revenue Ratio 2/								
Baseline	9	9	8	8	7	7	5	4
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	9	9	8	9	10	11	11	13
A2. Primary balance is unchanged from 2013	9	9	8	8	8	8	6	8
A3. Permanently lower GDP growth 1/	9	9	8	8	8	8	6	6
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	9	9	9	9	8	9	7	9
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	9	9	8	9	8	8	7	5
B3. Combination of B1-B2 using one half standard deviation shocks	9	9	9	9	9	9	8	9
B4. One-time 30 percent real depreciation in 2014	9	10	10	10	10	10	8	7
B5. 10 percent of GDP increase in other debt-creating flows in 2014	9	9	9	9	8	8	7	5

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.